

PETER MIRITHU

Software Engineer | AI/ML Engineer | Mobile App Developer

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Portfolio: petermirithu.github.io/Portfolio | Location: Nairobi, Kenya

ABOUT ME

I am a Software Engineer with over 4 years of experience specializing in building intelligent, scalable systems across AI, full-stack web, and mobile development. I focus on turning complex problems into practical, production-ready solutions using Python, JavaScript, and modern AI tools like LangChain and LLM APIs, with experience spanning backend systems, mobile applications, machine learning, and cloud platforms such as AWS, GCP, and Azure. I am driven by designing and delivering impactful products especially in healthcare, education, and productivity where engineering meets real-world problem solving.

DIGITAL SKILLS

Programming Languages

• Python • JavaScript • TypeScript • SQL • PHP (basic)

Frontend Development

• React.js • React Native • Expo • EAS Build & Deploy • Angular • Electron.js • HTML5 • CSS3

Backend Development

• Django • Flask • FastAPI • Node.js • REST APIs

AI / Machine Learning / LLMs

• LangChain • LangGraph • OpenAI API • Google Gemini API • ChromaDB • Scikit-learn • OpenCV
• Machine Learning (Supervised Learning, Random Forest) • AI Agents / Multi-Agent Systems

Databases

• MongoDB • PostgreSQL • MySQL • ChromaDB • Firebase

Cloud & DevOps

• AWS • GCP • Microsoft Azure Foundry / AI services exposure • Vercel • Basic CI/CD workflows
• Server Deployment • Cloud Debugging & Production Troubleshooting

APIs & Integrations

• Mpesa API (payments integration) • Firebase Authentication • Cognito (AWS authentication flows)

Soft Skills / Professional Skills

• Problem Solving • Team Leadership • Objective-Oriented Execution • Adaptive Learning • Product Thinking
• Cross-functional Collaboration • Technical Communication

Languages

• English • Swahili

WORK EXPERIENCE

Aug 2024 – Current

Nairobi, Kenya

Software Engineer - Full Time

Vitafluence.ai

- Developed AI-powered agents for Lakesai using Python, Django, and Langchain, enabling seamless communication with external databases, accelerating research workflows.
- Designed a Windows desktop extraction software running Python scripts to process data collected by Neurobee, facilitating data analysis.
- Currently developing and training a gaze tracker machine learning model using Python, OpenCV, and Random Forest algorithm to enhance diagnostic accuracy.
- Resolved critical deployment challenges by creating a temporary AWS server to route traffic while troubleshooting Google Cloud Platform issues, ensuring uninterrupted service.
- Collaborate closely with cross-functional teams, including AI model developers and designers, to deliver visually appealing, AI-driven mobile and web interfaces.
- Deploy and manage scalable applications on cloud platforms such as AWS and Google Cloud, leveraging Python and React Native for front-end and back-end development.

May 2024 – Jul 2024

Frankfurt am Main,
Germany

Software Engineer - Intern

Vitafluence.ai

- Developed a teaching assistant AI agent for Lakesai, including an internal mobile app version, enhancing learning support for students and researchers.
- Created integration for AI-powered note-taking functionality similar to Notion, improving user productivity.

- Fixed critical issue allowing users to stop generative AI text mid-response, significantly improving user control and product experience.
- Built and deployed Xaidi, an AI health companion for the neurodivergent community, including mobile apps (React Native), backend servers (Python Django, Node.js), and AI models; deployed on Google Play Store, TestFlight, and AWS.
- Added multi-agent support to Xaidi, enabling simultaneous responses to multiple queries, greatly enhancing app capabilities.
- Represented Vitafluence.ai at VivaTech 2024 in Paris, showcasing Xaidi and emotion recognition AI models, promoting company innovations.
- Collaborated cross-functionally with diverse teams to develop AI health solutions, primarily using React Native, Python Django, Node.js, MongoDB, and ChromaDB for AI embeddings.
- Gained proficiency in React.js while working on Lakesai frontend, enhancing front-end development skills.

Oct 2020 – Apr 2024
Nairobi, Kenya

Software Engineer - Part Time

iZola LTD

- Enabled multi-language support (English/Swahili) for the Izuri research application, improving accessibility for diverse users.
- Implemented audio capture functionality and resolved bugs promptly to ensure smooth operation of the Izuri web app.
- Created and maintain Neurobee, an Android tablet app aiding early diagnosis of selective mutism and autism spectrum disorder in children, supporting ongoing research through regular updates and issue resolution.
- Developed client and therapist pages for the iZola Therapeutic Platform, creating APIs, managing databases, and ensuring intuitive UI/UX using Angular, Node.js, and MongoDB.
- Built an AI chatbot for user support on the Therapeutic Platform, enhancing client and therapist experience.
- Designed and managed an administrative dashboard for user authentication, session tracking, and revenue monitoring, streamlining therapist-client management.
- Represented iZola LTD at community events as a technical expert, addressing technical queries and assisting users with the Therapeutic Platform.
- Successfully balanced multiple projects by prioritizing tasks and managing time effectively to meet deadlines and deliver quality results.

PROJECTS

Mali Mobile App

AI-Powered Kenyan Financial Intelligence App | Ongoing

Mali is an AI-powered mobile app built for the Kenyan market that tracks real-world economic events fuel prices, food costs, and forex fluctuations and translates them into clear, personalised insights on what they mean for a user's daily cost of living and business. The app connects live news, prices, and macroeconomic data with AI reasoning to surface impact in simple, actionable terms.

Tech Stack: • React Native (Expo) • FastAPI • Firebase • LangChain • LangGraph • Supabase • Azure OpenAI

EDUCATION AND TRAINING

May 2020 – Aug 2024
Nairobi, Kenya

B.Sc. Applied Computer Technology

United States International University - Africa

Field: Computer Software Engineering | GPA 3.621, Cum Laude honors

CERTIFICATIONS

Apr 2025

Nairobi, Kenya

Certificate of Achievement

AI Campus – The Learning Platform for Artificial Intelligence

Aug – Oct 2024

Nairobi, Kenya

Certificate of Recognition

A2SV | Africa to Silicon Valley

Aug 2019 – Jan 2020

Nairobi, Kenya

Certificate of Excellence in Web Development

Moringa School

Apr – Jun 2019

Kalungu, Zambia

Certificate of Attendance in Discipleship Mission Training

Foundation for Cross-cultural Education

Jan – Apr 2019

Nairobi, Kenya

Certificate of Participation in ICT Competence

Daystar University